

Yuma Mizuno

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Education

- Mar 2021 **Doctor of Science**, *Department of Mathematical and Computing Science, Tokyo Institute of Technology*
Thesis: Difference equations arising from cluster algebras
Supervisor: Yuji Terashima and Sakie Suzuki
- Mar 2018 **Master of Science**, *Department of Mathematical and Computing Science, Tokyo Institute of Technology*
Supervisor: Yuji Terashima
- Mar 2016 **Bachelor of Science**, *Department of Information Science, Tokyo Institute of Technology*

Teaching

- Semester 2, 2025–2026 Lecturer, University College Cork, Measure Theory and Integration (MA3064)
- Fall 2021 Part-time lecturer, Tokyo University of Agriculture and Technology, Calculus II and Exercises

Research Positions and Fellowships

- 2025–present Postdoctoral Researcher, University College Cork
- 2024–2025 Postdoctoral Researcher, University College Dublin
- 2021–2024 JSPS Research Fellow (PD), Japan Society for the Promotion of Science
- 2018–2021 JSPS Research Fellow (DC1), Japan Society for the Promotion of Science

Preprints

- Y. Mizuno. *Rotation fixed points of undecorated braid varieties and fusion rings of affine Lie algebras*. Oct. 1, 2026. arXiv: 2610.02034 [math.RT].
- Y. Mizuno. *The three Kanade-Russell identities modulo nine*. Sept. 9, 2026. arXiv: 2609.09653 [math.NT].
- Y. Mizuno. *Mutation Sequences along Weaves and Amalgamation of Braid Varieties*. Sept. 5, 2026. arXiv: 2609.05833 [math.AG].
- T. Ishibashi and Y. Mizuno. *Adjoint Reidemeister torsion of 3-manifolds with torus boundary for semisimple algebraic groups*. 2026. arXiv: 2603.00816 [math.GT].
- M. de Courcy-Ireland, M. Litman, and Y. Mizuno. *Divisibility by p for Markoff-like Surfaces*. 2025. arXiv: 2509.02187 [math.NT].

Publications

- S. Sonoda, K. Kasaura, Y. Mizuno, K. Tsukamoto, and N. Onda. *Lean Formalization of Generalization Error Bound by Rademacher Complexity and Dudley's Entropy Integral*. 17th International Conference on Interactive Theorem Proving (ITP 2026). Vol. 382. Leibniz International Proceedings in Informatics (LIPIcs). July 16, 2026, 8:1–8:17. DOI: 10.4230/LIPIcs.ITP.2026.8.
- Y. Mizuno. *Periodic Y -systems and Nahm sums: the rank 2 case*. SIGMA Symmetry Integrability Geom. Methods Appl. 21, 094 (2025). DOI: 10.3842/sigma.2025.094.

- Y. Mizuno. *Remarks on Nahm sums for symmetrizable matrices*. Ramanujan Journal 66, 62 (2025). DOI: 10.1007/s11139-025-01033-6.
- Y. Mizuno. *q -Painlevé equations on cluster Poisson varieties via toric geometry*. Selecta Mathematica. New Series 30, 19 (2024). DOI: 10.1007/s00029-023-00906-2.
- Y. Mizuno. *Difference equations arising from cluster algebras*. Journal of Algebraic Combinatorics 54.1 (2021), pp. 295–351. DOI: 10.1007/s10801-020-00978-9.
- Y. Mizuno. *Exponents associated with Y -systems and their relationship with q -series*. SIGMA Symmetry Integrability Geom. Methods Appl. 16, 028 (2020). DOI: 10.3842/sigma.2020.028.
- Y. Mizuno. *Jacobian matrices of Y -seed mutations*. Advances in Applied Mathematics 115, 101987 (2020). DOI: 10.1016/j.aam.2019.101987.
- A. Kato, Y. Mizuno, and Y. Terashima. *Quiver mutation sequences and q -binomial identities*. International Mathematics Research Notices 2018.23 (2018), pp. 7335–7358. DOI: 10.1093/imrn/rnx108.

Invited Talks

- 13 Jul 2026 **Nahm sums from cluster transformations**, *Workshop on Higher Teichmüller theory, knot theory and cluster algebras*, Cork, Ireland.
- 18 May 2026 **Adjoint Reidemeister torsion for G -local systems on 3-manifolds with torus boundary**, *Paris algebra seminar*, Paris, France.
- 15 Apr 2026 **Adjoint Reidemeister torsion for G -local systems on 3-manifolds with torus boundary**, *Research Seminar on Quantum Topology and Categorification (QTCat)*, Hamburg, Germany.
- 17 Feb 2026 **Formalizing mathematics with Lean theorem prover**, *Boole Colloquium in Mathematics*, Cork, Ireland.
- 9 Dec 2025 **Adjoint Reidemeister torsion for G -local systems on 3-manifolds**, *Geometry and Mathematical Physics seminar*, Birmingham, UK.
- 13 Nov 2025 **Formalizing mathematics with theorem provers**, *The 28th Workshop on Information-Based Induction Sciences (IBIS2025)*, Naha, Japan.
- 31 Oct 2025 **Introduction to Lean theorem prover**, *iTHEMS Math & Computer Seminar*, Wako, Japan.
- 18 Jun 2025 **Lean theorem prover and monoidal category theory**, *BIMSA-Tsinghua Quantum Symmetry Seminar*, Online.
- 19 Feb 2025 **Geometry of q -Painlevé equations and cluster structures**, *Asymptotic Expansion of τ -functions and Related Topics*, Kyoto, Japan.
- 12 Mar 2024 **Cluster algebras and q -series**, *q -Series and Related Topics*, Osaka, Japan.
- 11 Mar 2024 **Cluster structures on q -Painlevé systems via toric geometry**, *Advances in Cluster Algebras 2024*, Nagoya, Japan.
- 16 Dec 2023 **Geometry of q -Painlevé systems and cluster structures**, *Comprehensive Studies of Differential Equations*, Tokyo, Japan.
- 21 Jun 2023 **Classification of periodic Y -systems of rank 2**, *Cluster algebras and related topics in East Asia*, Seoul, South Korea.
- 30 May 2023 **Modularity of Nahm sums and periodicity phenomena in cluster algebras**, *South Osaka Algebra Seminar*, Osaka, Japan.
- 26 Oct 2022 **q -Painlevé systems, toric surfaces, and cluster Poisson varieties**, *Painlevé Equations: From Classical to Modern Analysis*, Strasbourg, France.
- 14 May 2022 **Nahm's problem and cluster algebras**, *AMS Spring Western Virtual Sectional Meeting*, Online.

- 18 Nov 2021 **Nahm's problem and cluster algebras**, *UCD Algebra and Number Theory seminar*, Online.
- 28 Sep 2021 **q -Painlevé systems on cluster Poisson varieties**, *Infinite Analysis 21 Workshop Around Cluster Algebras*, Online.
- 17 Jul 2020 **T -systems and Y -systems in cluster algebras**, *South Osaka Algebra Seminar*, Online.
- 18 Jun 2020 **Difference equations arising from cluster algebras**, *Representation Theory Seminar (RIMS)*, Online.
- 12 Dec 2019 **The Cartan–Killing classification and its extensions in cluster algebras**, *Workshop on Combinatorics and Related Topics in Shinshu, Winter 2019*, Matsumoto, Japan.
- 13 Feb 2019 **Exponents associated with Y -systems**, *Tokyo Institute of Technology Topology Seminar*, Tokyo, Japan.
- 22 Jan 2019 **Combinatorics of ideal triangulations and quiver mutations**, *The 14th East Asian Conference on Geometric Topology*, Beijing, China.
- 6 Dec 2017 **Ideal triangulations of 3-manifolds and mutation loops**, *Infinite Analysis 17 -Algebraic and Combinatorial Aspects in Integrable Systems-*, Osaka, Japan.

Other Talks and Presentations

- 17 Jan 2025 **Metaprogramming on monoidal categories**, *Lean Together 2025*, Online.
- 30 Oct 2023 **Computations with morphisms in monoidal categories in Lean**, *The 19th Theorem Proving and Provers meeting*, Tokyo, Japan.
- 22 Oct 2022 **Computations on monoidal categories in the proof assistant Lean**, *Topology and Computers 2022*, Higashihiroshima, Japan.
- 14 Sep 2022 **Mutations of blowups of toric surfaces and q -Painlevé systems**, *Mathematical Society of Japan Autumn Meeting 2022*, Sapporo, Japan.
- 11 Aug 2022 **Mutations of blowups of toric surfaces and q -Painlevé systems**, *Summer Seminar on Functional Equations 2022*, Mori, Japan.
- 31 Mar 2022 **Mutations of blowups of toric surfaces and q -Painlevé system**, *Mathematical Society of Japan Annual Meeting 2022 (abstract only; no oral presentation)*, Online.
- 25 Jun 2021 **Cluster algebras and spaces of initial conditions for q -Painlevé systems**, *ALReT 2021*, Online.
- 4 Mar 2021 **q -Painlevé systems on cluster varieties**, *The 17th Mathematics Conference for Young Researchers*, Sapporo, Japan.
- 17 Jun 2019 **Exponents associated with periodic Y -systems**, *Cluster Algebras 2019 (CA19) (poster)*, Kyoto, Japan.
- 19 Mar 2019 **Exponents associated with Y -systems of type X_r at level l** , *Mathematical Society of Japan Annual Meeting 2019*, Tokyo, Japan.
- 6 Mar 2019 **Determinant identities associated with mutation sequences**, *The 15th Mathematics Conference for Young Researchers*, Sapporo, Japan.
- 24 Sep 2018 **Jacobian matrices of Y -seed mutations and mutation networks**, *Mathematical Society of Japan Autumn Meeting 2018*, Okayama, Japan.
- 13 Feb 2018 **Jacobian matrices of cluster transformations and representations of affine Lie algebras**, *The 1st Seminar for New Researchers in Mathematics*, Kyoto, Japan.
- 26 Mar 2017 **Quiver mutations and q -binomial identities**, *Mathematical Society of Japan Annual Meeting 2017*, Hachioji, Japan.